Project Title: AI- Driven Phishing Email Detection Using NLP (Natural Language Processing)

Objective

To design and implement a machine- learning system that automatically detects phishing emails using text analysis and metadata features. Students will build the project **from scratch**, applying data preprocessing, feature extraction, model training, and evaluation.

Scope

Collect and preprocess email datasets (Kaggle, UCI, or custom samples).

Extract linguistic and structural features (URLs, sender domains, word patterns).

Train and compare models: Logistic Regression, Random Forest, Naive Bayes, and simple Neural Network.

Evaluate performance using accuracy, precision, recall, and F1- score.

Visualize confusion matrices and feature importance.

Methodology

Phase	Description
Data Collection	Gather phishing and legitimate email samples.
Data Cleaning	Remove HTML tags, punctuation, and stopwords.
Feature Engineering	Apply TF- IDF, word embeddings, and metadata extraction.
Model Development	Train multiple classifiers and tune hyperparameters.
Evaluation	Compare models and interpret results.
Deployment (Optional)	Create a simple web interface for live email classification.

Expected Deliverables

Python notebook with complete code and documentation.

Dataset (raw and cleaned).

Comparative analysis report.

Presentation slides summarizing findings and recommendations.

Learning Outcomes

Understand how NLP supports cybersecurity.

Gain hands- on experience in text classification and model evaluation.

Learn ethical AI practices for digital threat detection.

Strengthen teamwork and research presentation skills.

Suggested Tools

Libraries: scikit- learn, pandas, NLTK, spaCy, matplotlib, seaborn.

Environment: Jupyter Notebook or Google Colab.

Optional: Flask or Streamlit for deployment.

NOTE:

Natural Language Processing (NLP) is a branch of AI that enables computers to understand, interpret, and generate human language. It combines **linguistics** (grammar, semantics) with **machine learning** to process text and speech.

Why NLP in Cybersecurity?

Phishing emails, spam, and malicious messages often rely on **language tricks** — suspicious wording, fake urgency, or embedded URLs. NLP helps detect these patterns by:

Tokenizing text (breaking into words).

Extracting features (frequency of words, presence of URLs, sender domain).

Classifying messages as phishing vs. legitimate using ML models.

Example Workflow

1. **Preprocess text:** Remove HTML tags, punctuation, stopwords.
2. **Feature extraction:** Use TF- IDF or word embeddings to represent text numerically.
3. **Model training:** Apply Logistic Regression, Random Forest, or Neural Networks.
4. **Evaluation:** Measure accuracy, precision, recall, F1- score.

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